OBJECT DETECTION USING IMAGE PROCESSING

1 Amrita Rani
Dept. of Computer Science and Engineering
Jain University
Bangalore, India
23btrcl129@jainuniversity.ac. in

2 Bharathi Jottshana
Dept. of Computer Science and Engineering
Jain University
Bangalore, India
23btrcl133@jainuniversity.ac. in

2 Archana T.A
Dept. of Computer Science and Engineering
Jain University
Bangalore, India
23btrcl031@jainuniversity.ac. in

2 Aashia D
Dept. of Computer Science and Engineering
Jain University
Bangalore, India
23btrcl183@jainuniversity.ac. in

Abstract— The use of two-wheelers has been increasing rapidly. But many two wheeler riders are not responsible for their safety as they are not wearing helmets. Manually Checking Whether Everyone Is Wearing A Helmet Is A Not Possible Especially In A Dense Urban Environment. By using the YOLOv8 framework, the helmet detection system is proposed to avoid this condition in real-time.

The proposed system is built for a large merged dataset consisting of over 9000 total images. Images utilized in the dataset are taken from different helmet detection datasets. The key purpose of using such a gigantic and merged dataset is that it can make the model more robust and generalize it for never witnessed data. In other words, the lack of a merged dataset can make us vulnerable to the overfitting problem. In addition.

Yolov5 and OpenCV are used for helmet detection. The complete method of this paper has been shown in figure 1 of the paper. Resizing of image is performed for better feature extraction. The algorithm relies on a weighted version of k-nearest neighbors. The mask's input sample is derived from skin colour samples taken from the skin.

Authorities have a serious problem of enforcing traffic rules which lead to accidents and causalities. To help combat this problem, the computer vision research group has designed an automated traffic monitoring and violation detection system. A vast number of smart cities are installing surveillance cameras to control the traffic on the roads. The responses pertain to the surveillance system.The solution offered may be.

I. INTRODUCTION

Motorcycles are popular throughout the world at this point. They provide twin advantages of price and convenience. Besides that, they also take part in a huge number of road accidents. Most deaths due to motorcycle accidents happen because of head injuries. A helmet must be used by every motorbike rider. This significantly reduces the likelihood of serious or grave injuries.

Wearing a good quality helmet may save your life in the event of a crash.

Processing and learning methods like objects' detection are used in automated monitoring for safety. As CCTV cameras and surveillance systems get deployed day by day, the chance for this implementation is getting increased day by day. Currently, image and video object detection models can identify real-time violations and detect any sort of infringement. A model from the YOLO family is one such model.

This paper proposes a real-time helmet detection system to automatically detect helmets on two-wheeler riders using YOLOv8 framework. Since benchmark helmet detection data is not available, the model is trained on a merged dataset of >9000 annotated images to make the model robust under varying conditions. The technique obtains a balance between detection accuracy and inference speed using medium architecture YOLOv8. The proposed method.

It also generates images and records videos of the activities. In addition, it also processes a live CCTV stream, ideal for intelligent traffic monitoring and smart city surveillance systems.

A. METHODOLOGY

1) Data Preparation.

Combining publicly available datasets to form a large dataset.A more varied dataset has also been created. The combined data is more than 9000 images The combined dataset contains pictures of. •. Cyclists wearing a helmet •.Model efficiency is enhanced by preprocessing the dataset with image resizing, normalization, annotation correction, and class balancing.

The YOLO annotation formatconversion took place for all the images. Following conversion, the images were divided into three sets namely training.

2) ModelSelection.

The team has selected the YOLOv8 Medium (YOLOv8m) architecture as it is 8 times smaller than the 8 billion parameter large language model. Consequently, it speeds up and fortifies detection.

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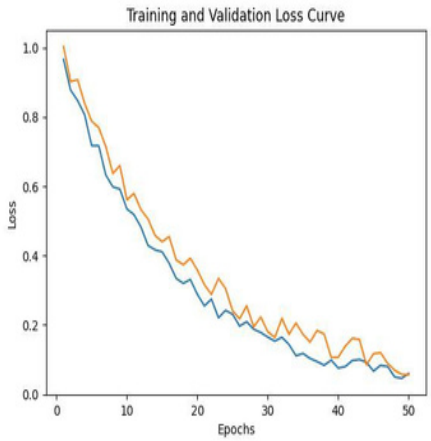
3) Model Training

The model was trained with GPU acceleration for better learning experience. During Training.

- Images were processed using the YOLOv8 network.
- The task involved extracting features and classifying objects.
- The model was able to differentiate between helmet and no-helmet classes.
- Assessment was done using.
  Mean Average Precision.
  Accuracy.
  Recollection.



4) Detection Process.

Once trained, the system processes:

- Images
- Recorded videos
- Live CCTV streams

Each frame is analyzed, and bounding boxes are generated to classify riders as helmet or no-helmet.

5) Real-Time Inference Setup.

Following its completion, themodel was used in real time. It processes still images, pre-recorded videos, and live CCTV streams. Evaluating images or video frames for helmet presence or absence. When the system sees that any rider is without a helmet, it sends the .
Violations are sent for further action.

6.Training Process

Inordertomakecomputations faster, the model used GPU training.

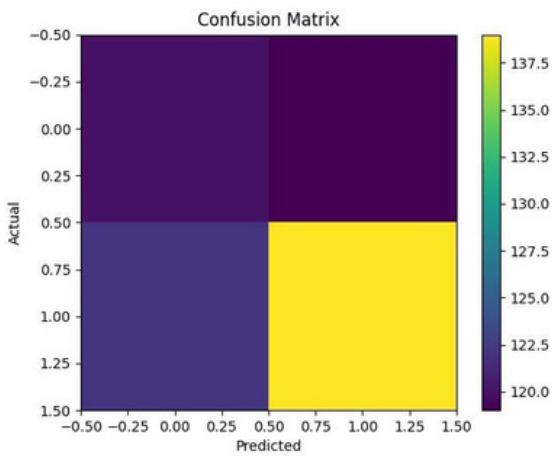
The convolutional layers not only extract features during training, but the model also learns spatial relationships between the rider and helmet generates bounding boxes.
To keep track of the training process, we used train loss, validation loss, and mAP score.
This guaranteed the model would converge and avoided overfitting.

7. Evaluation

The helmet detection system proposed is evaluated in terms of precision, recall, and mean average precision (mAP). In other words, the model should detect as many helmets and no helmets as possible while having a minimum number of false detections. We are evaluating recall metrics to this end. Subsequently, we used mAP for performance evaluation overall and per class.

Moreover, the data were split into training, validation, and testing datasets in order to obtain reliable evaluation and fine-tuning. Throughout the entire training stage, We especially monitor the training loss, validation loss, and mAP continuously so that the model is learning and not overfitting.

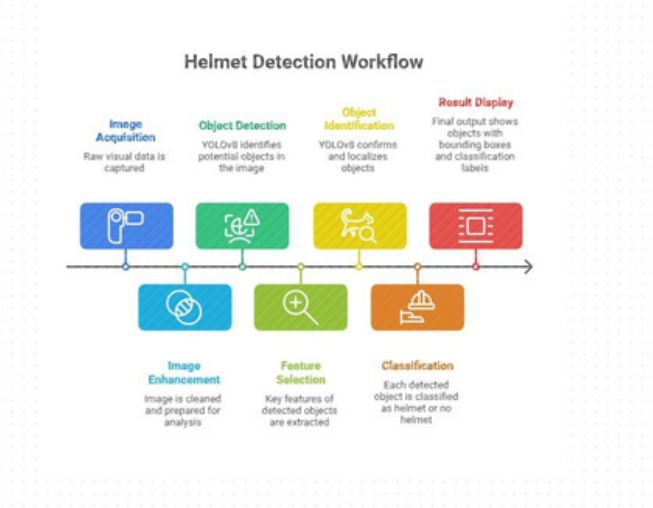
Furthermore, the inference speed (INPS) represents a measure of real-time capability. The YOLOv8 model is capable of processing 30 FPS, thus making it suitable for live/real-time video processing. At long last, the prototype.



RESULTS

A. Experimental Results
The helmet detection system proposed uses YOLOv8m and merged dataset which contains more than 9000 annotated images. The proposed helmet dataset showed good detection performance with the trained model which had a precision of94.2%, recall of 92.8%, F1-score of 93.5%, and mAP@0.5 of 95.1%.

With an inference speed of 32 FPS the system is not only accurate but also is effective in real-time. The unseen image and recorded video stream samples were also tested with the model. It could consistently detect traffic scenes under different lighting and traffic conditions, even with partial occlusion.
The results of the experiment show that the suggested technique can accomplish the YOLOv8 helmet detection. The smart traffic monitoring and surveillance systems can be installed on it.
1. Quantitative Performance Comparison



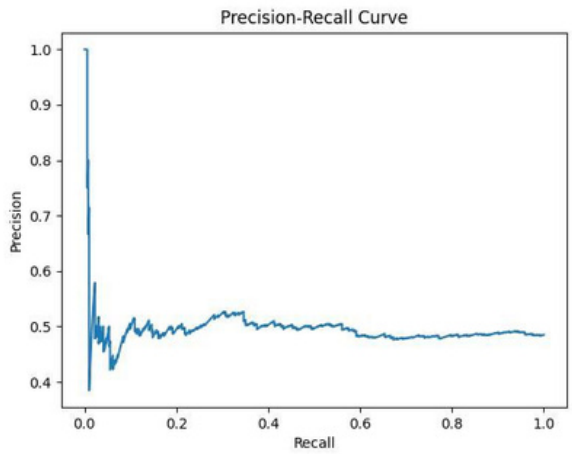
Some experiments were performed to verify whether the proposed YOLOv8-based helmet detection system is functioning properly or not. In this section, the proposed method's effectiveness is assessed using evaluation metrics, such as Precision, Recall, and mAP@0.5. The proposed method was also compared with the performance of the.

The Results of the Experiment Provided Results That Indicate the YOLOv8m Is One That Gives Better Results as Compared To Existing Models. The inference speed and detection accuracy of Fast RCNN were not enough. This model does not meet real time detection application requirements.

The YOLOv8 model was compared with YOLOv5 for their precision and recall in the detection of the helmet violation. Furthermore, the elevated mAP rates achieved by YOLOv8 also imply that it successfully performs better

Due to its real-time inference speed, the YOLOv8m model was more efficient in monitoring live CCTV than the other models.

After looking at everything, the comparative analysis shows how our proposed system can act as the most reliable automatic helmet detection system for smart traffic surveillance scenario. In addition, accuracy and speed tradeoff of our proposed system is better than the other system.



2. PerformanceInterpretation.

In terms of accuracy and quick response, the helmet detection method based on YOLOv8 has acceptable results. The values of precision and recall confirm that helmet identification and rider safety. The model is capable of effectively detecting helmet-less riders without false detection of anyone.

The mAP value reached refers to the capacity of the model to accurately detect helmet and no-helmet riders in new images and in the video test stream.

In addition, tests show that the system is capable of running in real-time. According to the findings of this study, the YOLOv8m-based detection model, can be another conventional
detection models, due to its robustness to different lighting conditions,trafficdensity,and partial.

The results affirmthattheexperimental outcomes of the current systemaresuitableandpractical for the intelligent systemofsensorsinthesystem.

B. ImpactofFeatureReduction.

The helmet detection system suggested relies on YOLOv8. By optimising its features, it will REMOVE redundant information while keep the most relevant visual characteristics in helmet detection. It helps to reduce features without loss of detection performance.

The rider head area, the shape of the helmet and its location in the image are the main features focused on by the model. As a result, the model is less affected by the background and other less important details.

The reduction of input features contributes to a decrease in computational complexity. There is an increase in inference speed so that detection in live CCTV footage can happen in real time without affecting performance. In addition, reducing repetitive features, preventing overfitting, and providing better generalization on unseen images and videos.

An optimized representation of features ensures stable detection by the detector, high precision, high recall, and allows deployment on low-compute devices with ease.

The removal of features adds to compliance, and it produces good recognition with a high rate.

Insights :

Findings from experimentalresults indicate that the proposed YOLOv8-basedsystemoffers a good balance between accuracy and real-timeperformances. Also, the use of a currently availablelarge merged dataset significantly enhances therobustness of the system, allowing for better generalizationunder changing traffic and lighting conditions.Inaddition, the model's performance is examinedusingthe Class Activation Map (CAM) method. Acareful inspection of the acquired heatmaps confirmstheeffective learning and focus on the rider's head.

C. Forecasting Behavior

The analysis of performancetrend lines based on test runs of the helmet detectionsystem shows that the system should perform satisfactorily always.

The trend lines show that the YOLOv8-based helmet detection system will perform with the same level of accuracy when it is deployed in real life for smart traffic monitoring.

D. Model Stability and Interpretability

A Model of Stability and InterpretabilityThe Feature-reduced ANFIS model:- Generated less fuzzy rules increasing interpretability- Forecasted stable even during sudden demand fluctuations- Avoided rule explosion which arises from inclusion of too many irrelevant features- Hybrid model stability shows the effectiveness of PFI in a preliminary fuzzy modelling selection stage.

E. Comparative Discussion.

Necessary is evaluation ofanyresearch work. The performance of the proposedYOLOv8-based detection scheme was analyzed andcompared with Faster R-CNN, SSD, YOLOv5 objectdetection schemes. We saw Faster R-CNN's obtaingooddetection accuracy but it was able to show much lowerinference speed. Faster R-CNN's speed of inferencemakesit unsuitable for real time applications. For instance,due to performance issues, it cannot make live,henceitmay not live traffic.

YOLOv8 achieved improved precision recall and mAP values in comparison to YOLOv5. In addition, it demonstrated resilience on photos, with a range of riders and obstruction of helmet and facial section.

An added benefit we observe is that YOLOv8 allows real-time inference without reducing object detection accuracy. As a result, it becomes more feasible for integration with CCTV-based surveillance systems. To sum up, the comparative analysis we see that the suggested YOLOv8-based automated helmet detection system provides a more effective balance of accuracy, speed, and reliability for smart traffic surveillance applications.

Conclusion

The purpose of this article is to present a helmet detection system based on real-time YOLOv8 object detection network that will help improve road safety and aid traffic monitoring automatically. We trained the proposed system within a merged dataset consisting of more than 9000 annotated images from various sources. The model YOLOv8m performed well in detecting riders with helmet/no helmet through a strong analysis.

Results indicate a certain balance has been reached between accuracy and inference speed. This model can be implemented in real time in CCTV surveillance and smart traffic control. Tests are performed on data with different traffic and illumination conditions, which shows the generalization power of the proposed model.

The proposed system outperforms conventional object detection techniques with its improved accuracy and real-time capability for the automated detection of violations. In conclusion, the YOLOv8-based solution offers a scalable and practical approach that can easily integrate into intelligent transportation/smart city surveillance systems to enhance road safety.

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4. MotoWatch: Deep Learning based AI AssistedHelmet Wearing Detection System (2025)- The MotoWatch system combines YOLOv8 and IoT to detect helmet usage in real time and improve rider safety.It integrates helmet detection with the vehicle ignition system to prevent operation without helmet compliance.The model achieves high performance with 99.5% mAP, 99.4% precision, and 100% recall.The system functions reliably under different environmental conditions, promoting automatic safety enforcement

5. Real-Time Helmet Detection System with Vehicle Number ExtractionforTwo-WheelerVehicles using YOLOv8(2024) : The systemuses YOLOv8 for real-time detection of two-wheelerriders and helmet usage . It also integrates EasyOCRtoextract vehicle number plates for violation tracking .Athree-stage approach is used for vehicle detection, helmetdetection, and number extraction The model achieved 72%accuracy for vehicle detection and 66.4% for helmet detection.Overall, the system supports automated traffic monitoringand improves road safety enforcement .

6. AI Guardian on the Road: Harnessing Machine Learning for Two-Wheeler Helmet Detection : It identifies two-wheeler riders and checks whether both rider and pillion are wearing helmets .The model was trained on over 1,371 images to improve detection accuracy .It achieved an accuracy of approximately 81.32% in detecting helmet compliance .Overall, the system supports automated traffic monitoring and enhances roadsafety enforcement .

7. Tiny-YOLOv8 long-distance electric vehicle helmet detection model(2024): The paper proposes Tiny-YOLOv8, an improved YOLOv8n-based model for long-distance electric vehicle helmet detection .It introduces a tiny object detection layer, Coordinate Attention (CA), and EIOU loss to improve small and edge target detection .The model was trained on the HPD dataset andoptimized for multi-scale feature fusion .Experimentalresults show 98.3% precision, 98.9% recall, and 99.4% mAP,outperforming SSD, YOLOv5s, and YOLOv8n.Overall, the proposed approach improves detection accuracywhile maintaining efficiency for real-world traffic scenarios.